Guru Nanak Dev Engineering College, Ludhiana

Department of Computer Science & Engineering

LABORATORY FILE

Artificial Intelligence

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Class : Second Year — Div C2

Subject : Artificial Intelligence

Professor : Prof. Jasdeep Kaur Joia

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List of Practicals

Sr. No.	Experiment Title	Date	Sign	Marks
1	Implement a simple reflex agent for the vacuum-cleaner world.			
2	Implement BFS and DFS for water jug problem.			
3	Implement Hill Climbing search algorithm.			
4	Implement A* algorithm for eight puzzle problem.			
5	Develop Tic Tac Toe using Minimax with alpha-beta pruning.			
6	Represent a real-world domain using frames and semantic networks.			
7	Implement Goal Stack Planning for the Blocks World.			
8	Implement Bayes Rule to classify emails as spam or not spam.			
9	Build a small neural network from scratch.			
10	Generate images from text using a pre-trained diffusion/GAN model.			

Experiment No: 1 Mantej Singh Josan — 2435087
Date:

Implement a simple reflex agent for the vacuum-cleaner world.

Aim

A Simple Reflex Agent

Theory

A simple reflex agent selects actions based only on the current percept, ignoring all prior history. It uses condition-action rules of the form *if percept then action*. The vacuum-cleaner world consists of two locations (A and B), each of which can be dirty or clean. The agent perceives its current location and whether it is dirty, then acts accordingly — either cleaning or moving to the other location.

Task Requirement

Performance Measure	Environment	Actuators	Sensors
1. Energy efficiency (battery saved)	A room or grid of squares	Suction pump, wheels, motor	Dirt detector, location sensor, bump sensor
2. Time saved			
3. Cleanliness (percent of space clean)			

Code

A simple implementation using `if-else` that runs the simulation only for 1 episode.

```python
def reflex_agent(location, status):
      if status == "Dirty":
          return "Suck"
      elif location == "A":
          return "MoveRight"
      else:
          return "MoveLeft"

# Test the agent
world = [("A", "Dirty"), ("A", "Clean"), ("B", "Dirty"), ("B", "Clean")]

print(f" {'location':<8} | {'Status':<6} | Action")
print("-" * 35)
for loc, stat in world:
  action = reflex_agent(loc, stat)
  print(f" {loc:<8} | {stat:<6} | {action}")
```

```
 location | Status | Action
-----------------------------------
 A        | Dirty  | Suck
 A        | Clean  | MoveRight
 B        | Dirty  | Suck
 B        | Clean  | MoveLeft
```

Conclusion

The simple reflex agent was successfully implemented for the vacuum-cleaner world. The agent correctly determines its action based solely on the current percept without any memory of past states. This demonstrates the basic structure of a condition-action rule-based agent.

Date: Faculty Signature: _____

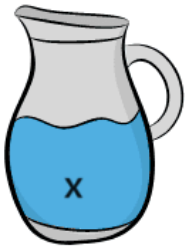
Experiment No: 2 Mantej Singh Josan — 2435087

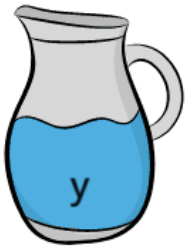
Date:

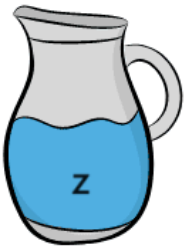
Implement BFS and DFS for water jug problem.

Aim

Water jug problem








```python
def main():
    print('hello water')
main()
```

```
hello water
```

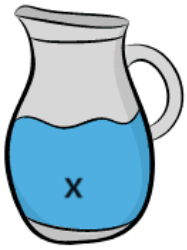
Date: Faculty Signature: _____

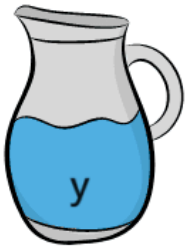
Experiment No: 3 Mantej Singh Josan — 2435087

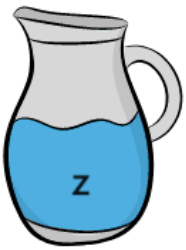
Date:

Implement Hill Climbing search algorithm.

Water Jug







Aim

IDK

```python
def main():
    print(f"{'yo':*^10}")
main()
```

```
****yo****
```

Date: Faculty Signature: _____